

Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z1,Z4,Z5

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Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple Z1 Z4 Z5: $\frac{Z_1 Z_4 Z_5 g_m - Z_1 Z_4}{2Z_1 Z_4 g_m + 2Z_1 Z_5 g_m + 2Z_1 + Z_4 + 2Z_5}$

$$H(z) = \frac{Z_1 Z_4 Z_5 g_m - Z_1 Z_4}{2Z_1 Z_4 g_m + 2Z_1 Z_5 g_m + 2Z_1 + Z_4 + 2Z_5}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

7.1 LP-1 $Z(s) = \left(\frac{1}{C_1 s}, \; \infty, \; \infty, \; \frac{1}{C_4 s}, \; R_5, \; \infty \right)$

$$H(s) = \frac{R_5 g_m - 1}{2C_1 C_4 R_5 s^2 + 2g_m + s \left(C_1 + 2C_4 R_5 g_m + 2C_4 \right)}$$

Parameters:

Q: $\frac{2\sqrt{C_1}\sqrt{C_4}\sqrt{R_5}\sqrt{g_m}}{C_1+2C_4R_5g_m+2C_4}$
 wo: $\frac{\sqrt{g_m}}{\sqrt{C_1}\sqrt{C_4}\sqrt{R_5}}$
 bandwidth: $\frac{C_1+2C_4R_5g_m+2C_4}{2C_1C_4R_5}$
 K-LP: $\frac{R_5g_m-1}{2g_m}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.2 LP-2 $Z(s) = \left(\frac{1}{C_1 s}, \; \infty, \; \infty, \; \frac{R_4}{C_4 R_4 s + 1}, \; R_5, \; \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4}{2C_1 C_4 R_4 R_5 s^2 + 2R_4 g_m + 2R_5 g_m + s \left(C_1 R_4 + 2C_1 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_4 \right) + 2}$$

Parameters:

Q: $\frac{2\sqrt{C_1}\sqrt{C_4}\sqrt{R_4}\sqrt{R_5}\sqrt{R_4g_m+R_5g_m+1}}{C_1R_4+2C_1R_5+2C_4R_4R_5g_m+2C_4R_4}$
 wo: $\frac{\sqrt{R_4g_m+R_5g_m+1}}{\sqrt{C_1}\sqrt{C_4}\sqrt{R_4}\sqrt{R_5}}$
 bandwidth: $\frac{C_1R_4+2C_1R_5+2C_4R_4R_5g_m+2C_4R_4}{2C_1C_4R_4R_5}$
 K-LP: $\frac{R_4R_5g_m-R_4}{2R_4g_m+2R_5g_m+2}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.3 LP-3 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1}{2C_1 C_4 R_1 R_5 s^2 + 2R_1 g_m + s(C_1 R_1 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_5}\sqrt{2R_1 g_m + 1}}{C_1 R_1 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5}$
 wo: $\frac{\sqrt{2}\sqrt{2R_1 g_m + 1}}{2\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_5}}$
 bandwidth: $\frac{C_1 R_1 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5}{2C_1 C_4 R_1 R_5}$
 K-LP: $\frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.4 LP-4 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2C_1 C_4 R_1 R_4 R_5 s^2 + 2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s(C_1 R_1 R_4 + 2C_1 R_1 R_5 + 2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4}\sqrt{R_5}\sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{C_1 R_1 R_4 + 2C_1 R_1 R_5 + 2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5}$
 wo: $\frac{\sqrt{2}\sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{2\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4}\sqrt{R_5}}$
 bandwidth: $\frac{C_1 R_1 R_4 + 2C_1 R_1 R_5 + 2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5}{2C_1 C_4 R_1 R_4 R_5}$
 K-LP: $\frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(R_1, \infty, \infty, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_5 s + R_1 R_5 g_m - R_1}{2C_4 C_5 R_1 R_5 s^2 + 2R_1 g_m + s(2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_1}\sqrt{R_5}\sqrt{2R_1 g_m + 1}}{2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5}$
 wo: $\frac{\sqrt{2}\sqrt{2R_1 g_m + 1}}{2\sqrt{C_4}\sqrt{C_5}\sqrt{R_1}\sqrt{R_5}}$
 bandwidth: $\frac{2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5}{2C_4 C_5 R_1 R_5}$
 K-LP: $\frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_1 R_5}{2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5}$
 Qz: None
 Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(R_1, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_4 s + R_1 R_4 g_m}{2C_4 C_5 R_1 R_4 s^2 + 2R_1 g_m + s(2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_4}\sqrt{C_5}\sqrt{R_1}\sqrt{R_4}\sqrt{R_1 g_m + 1}}{2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4} \\ \text{wo: } & \frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_4}\sqrt{C_5}\sqrt{R_1}\sqrt{R_4}} \\ \text{bandwidth: } & \frac{2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4}{2C_4 C_5 R_1 R_4} \\ \text{K-LP: } & \frac{R_1 R_4 g_m}{2R_1 g_m + 2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5 R_1 R_4}{2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.3 X-INVALID-NUMER-3 $Z(s) = \left(R_1, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_4 R_5 s + R_1 R_4 R_5 g_m - R_1 R_4}{2C_4 C_5 R_1 R_4 R_5 s^2 + 2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s(2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_1}\sqrt{R_4}\sqrt{R_5}\sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{2\sqrt{C_4}\sqrt{C_5}\sqrt{R_1}\sqrt{R_4}\sqrt{R_5}} \\ \text{bandwidth: } & \frac{2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5}{2C_4 C_5 R_1 R_4 R_5} \\ \text{K-LP: } & \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5 R_1 R_4 R_5}{2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.4 X-INVALID-NUMER-4 $Z(s) = \left(R_1, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 R_4 g_m + s(C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4)}{2R_1 g_m + s^2(2C_4 C_5 R_1 R_4 R_5 g_m + 2C_4 C_5 R_1 R_4 + 2C_4 C_5 R_4 R_5) + s(2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 R_5 g_m + 2C_5 R_1 + C_5 R_4 + 2C_5 R_5) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_4}\sqrt{C_5}R_1\sqrt{R_4}R_5g_m\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+2\sqrt{C_4}\sqrt{C_5}R_1\sqrt{R_4}\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+2\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}R_5\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}}{2C_4R_1R_4g_m+2C_4R_4+2C_5R_1R_4g_m+2C_5R_1R_5g_m+2C_5R_1+C_5R_4+2C_5R_5} \\ \text{wo: } & \sqrt{\frac{R_1g_m+1}{C_4C_5R_1R_4R_5g_m+C_4C_5R_1R_4+C_4C_5R_4R_5}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{R_1g_m+1}{C_4C_5R_1R_4R_5g_m+C_4C_5R_1R_4+C_4C_5R_4R_5}}(2C_4R_1R_4g_m+2C_4R_4+2C_5R_1R_4g_m+2C_5R_1R_5g_m+2C_5R_1+C_5R_4+2C_5R_5)}{2\sqrt{C_4}\sqrt{C_5}R_1\sqrt{R_4}R_5g_m\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+\frac{1}{R_1R_5g_m+R_1+R_5}+2\sqrt{C_4}\sqrt{C_5}R_1\sqrt{R_4}\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+2\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}R_5\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}} \\ \text{K-LP: } & \frac{R_1R_4g_m}{2R_1g_m+2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_5R_1R_4R_5g_m-C_5R_1R_4}{2C_4R_1R_4g_m+2C_4R_4+2C_5R_1R_4g_m+2C_5R_1R_5g_m+2C_5R_1+C_5R_4+2C_5R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.5 X-INVALID-NUMER-5 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, R_4, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_4 s + R_4 g_m}{C_1 C_5 R_4 s^2 + 2g_m + s(2C_1 + 2C_5 R_4 g_m + 2C_5)}$$

Parameters:

$$\text{Q: } \frac{\sqrt{2}\sqrt{C_1}\sqrt{C_5}\sqrt{R_4}\sqrt{g_m}}{2C_1 + 2C_5 R_4 g_m + 2C_5}$$

wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_4}}$
bandwidth: $\frac{2C_1+2C_5R_4g_m+2C_5}{C_1C_5R_4}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $-\frac{C_5R_4}{2C_1+2C_5R_4g_m+2C_5}$
Qz: None
Wz: None

8.6 X-INVALID-NUMER-6 $Z(s) = \left(\frac{1}{C_1s}, \infty, \infty, R_4, \frac{R_5}{C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{-C_5R_4R_5s + R_4R_5g_m - R_4}{C_1C_5R_4R_5s^2 + 2R_4g_m + 2R_5g_m + s(C_1R_4 + 2C_1R_5 + 2C_5R_4R_5g_m + 2C_5R_5) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_5}\sqrt{R_4}\sqrt{R_5}\sqrt{R_4g_m+R_5g_m+1}}{C_1R_4+2C_1R_5+2C_5R_4R_5g_m+2C_5R_5}$
wo: $\frac{\sqrt{2R_4g_m+2R_5g_m+2}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_4}\sqrt{R_5}}$
bandwidth: $\frac{\sqrt{2}\sqrt{2R_4g_m+2R_5g_m+2}(C_1R_4+2C_1R_5+2C_5R_4R_5g_m+2C_5R_5)}{2C_1C_5R_4R_5\sqrt{R_4g_m+R_5g_m+1}}$
K-LP: $\frac{R_4R_5g_m-R_4}{2R_4g_m+2R_5g_m+2}$
K-HP: 0
K-BP: $-\frac{C_5R_4R_5}{C_1R_4+2C_1R_5+2C_5R_4R_5g_m+2C_5R_5}$
Qz: None
Wz: None

8.7 X-INVALID-NUMER-7 $Z(s) = \left(\frac{1}{C_1s}, \infty, \infty, R_4, R_5 + \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{R_4g_m + s(C_5R_4R_5g_m - C_5R_4)}{2g_m + s^2(C_1C_5R_4 + 2C_1C_5R_5) + s(2C_1 + 2C_5R_4g_m + 2C_5R_5g_m + 2C_5)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_5}\sqrt{g_m}\sqrt{R_4+2R_5}}{2C_1+2C_5R_4g_m+2C_5R_5g_m+2C_5}$
wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1C_5R_4+2C_1C_5R_5}}$
bandwidth: $\frac{2C_1+2C_5R_4g_m+2C_5R_5g_m+2C_5}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_4+2R_5}\sqrt{C_1C_5R_4+2C_1C_5R_5}}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_5R_4R_5g_m-C_5R_4}{2C_1+2C_5R_4g_m+2C_5R_5g_m+2C_5}$
Qz: None
Wz: None

8.8 X-INVALID-NUMER-8 $Z(s) = \left(\frac{1}{C_1s}, \infty, \infty, \frac{1}{C_4s}, \frac{R_5}{C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{-C_5R_5s + R_5g_m - 1}{2g_m + s^2(2C_1C_4R_5 + C_1C_5R_5 + 2C_4C_5R_5) + s(C_1 + 2C_4R_5g_m + 2C_4 + 2C_5R_5g_m)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{R_5}\sqrt{g_m}\sqrt{2C_1C_4+C_1C_5+2C_4C_5}}{C_1+2C_4R_5g_m+2C_4+2C_5R_5g_m}$
wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{2C_1C_4R_5+C_1C_5R_5+2C_4C_5R_5}}$
bandwidth: $\frac{C_1+2C_4R_5g_m+2C_4+2C_5R_5g_m}{\sqrt{R_5}\sqrt{2C_1C_4+C_1C_5+2C_4C_5}\sqrt{2C_1C_4R_5+C_1C_5R_5+2C_4C_5R_5}}$
K-LP: $\frac{R_5g_m-1}{2g_m}$
K-HP: 0
K-BP: $-\frac{C_5R_5}{C_1+2C_4R_5g_m+2C_4+2C_5R_5g_m}$
Qz: None
Wz: None

8.9 X-INVALID-NUMER-9 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_4 s + R_4 g_m}{2g_m + s^2 (2C_1 C_4 R_4 + C_1 C_5 R_4 + 2C_4 C_5 R_4) + s (2C_1 + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{R_4}\sqrt{g_m}\sqrt{2C_1 C_4 + C_1 C_5 + 2C_4 C_5}}{2C_1 + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{g_m}}{\sqrt{2C_1 C_4 R_4 + C_1 C_5 R_4 + 2C_4 C_5 R_4}} \\ \text{bandwidth: } & \frac{2C_1 + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5}{\sqrt{R_4}\sqrt{2C_1 C_4 + C_1 C_5 + 2C_4 C_5}\sqrt{2C_1 C_4 R_4 + C_1 C_5 R_4 + 2C_4 C_5 R_4}} \\ \text{K-LP: } & \frac{R_4}{2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5 R_4}{2C_1 + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.10 X-INVALID-NUMER-10 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_4 R_5 s + R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + s^2 (2C_1 C_4 R_4 R_5 + C_1 C_5 R_4 R_5 + 2C_4 C_5 R_4 R_5) + s (C_1 R_4 + 2C_1 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_5) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{R_4}\sqrt{R_5}\sqrt{2C_1 C_4 + C_1 C_5 + 2C_4 C_5}\sqrt{R_4 g_m + R_5 g_m + 1}}{C_1 R_4 + 2C_1 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_5} \\ \text{wo: } & \frac{\sqrt{2R_4 g_m + 2R_5 g_m + 2}}{\sqrt{2C_1 C_4 R_4 R_5 + C_1 C_5 R_4 R_5 + 2C_4 C_5 R_4 R_5}} \\ \text{bandwidth: } & \frac{\sqrt{2}\sqrt{2R_4 g_m + 2R_5 g_m + 2}(C_1 R_4 + 2C_1 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_5)}{2\sqrt{R_4}\sqrt{R_5}\sqrt{2C_1 C_4 + C_1 C_5 + 2C_4 C_5}\sqrt{R_4 g_m + R_5 g_m + 1}\sqrt{2C_1 C_4 R_4 R_5 + C_1 C_5 R_4 R_5 + 2C_4 C_5 R_4 R_5}} \\ \text{K-LP: } & \frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + 2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5 R_4 R_5}{C_1 R_4 + 2C_1 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.11 X-INVALID-NUMER-11 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_5 g_m + s (C_4 R_4 R_5 g_m - C_4 R_4) - 1}{2g_m + s^2 (C_1 C_4 R_4 + 2C_1 C_4 R_5) + s (C_1 + 2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{g_m}\sqrt{R_4 + 2R_5}}{C_1 + 2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1 C_4 R_4 + 2C_1 C_4 R_5}} \\ \text{bandwidth: } & \frac{C_1 + 2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4}{\sqrt{C_1}\sqrt{C_4}\sqrt{R_4 + 2R_5}\sqrt{C_1 C_4 R_4 + 2C_1 C_4 R_5}} \\ \text{K-LP: } & \frac{R_5 g_m - 1}{2g_m} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_4 R_4 R_5 g_m - C_4 R_4}{C_1 + 2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.12 X-INVALID-NUMER-12 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_4 s + R_1 R_4 g_m}{C_1 C_5 R_1 R_4 s^2 + 2R_1 g_m + s (2C_1 R_1 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}\sqrt{R_4}\sqrt{R_1 g_m + 1}}{2C_1 R_1 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4} \\ \text{wo: } & \frac{\sqrt{2R_1 g_m + 2}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}\sqrt{R_4}} \end{aligned}$$

bandwidth: $\frac{\sqrt{2}\sqrt{2R_1g_m+2}(2C_1R_1+2C_5R_1R_4g_m+2C_5R_1+C_5R_4)}{2C_1C_5R_1R_4\sqrt{R_1g_m+1}}$
K-LP: $\frac{R_1R_4g_m}{2R_1g_m+2}$
K-HP: 0
K-BP: $-\frac{C_5R_1R_4}{2C_1R_1+2C_5R_1R_4g_m+2C_5R_1+C_5R_4}$
Qz: None
Wz: None

8.13 X-INVALID-NUMER-13 $Z(s) = \left(\frac{R_1}{C_1R_1s+1}, \infty, \infty, R_4, \frac{R_5}{C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{-C_5R_1R_4R_5s + R_1R_4R_5g_m - R_1R_4}{C_1C_5R_1R_4R_5s^2 + 2R_1R_4g_m + 2R_1R_5g_m + 2R_1 + R_4 + 2R_5 + s(C_1R_1R_4 + 2C_1R_1R_5 + 2C_5R_1R_4R_5g_m + 2C_5R_1R_5 + C_5R_4R_5)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}\sqrt{R_4}\sqrt{R_5}\sqrt{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}{C_1R_1R_4+2C_1R_1R_5+2C_5R_1R_4R_5g_m+2C_5R_1R_5+C_5R_4R_5}$
wo: $\frac{\sqrt{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}\sqrt{R_4}\sqrt{R_5}}$
bandwidth: $\frac{C_1R_1R_4+2C_1R_1R_5+2C_5R_1R_4R_5g_m+2C_5R_1R_5+C_5R_4R_5}{C_1C_5R_1R_4R_5}$
K-LP: $\frac{R_1R_4R_5g_m-R_1R_4}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}$
K-HP: 0
K-BP: $-\frac{C_5R_1R_4R_5}{C_1R_1R_4+2C_1R_1R_5+2C_5R_1R_4R_5g_m+2C_5R_1R_5+C_5R_4R_5}$
Qz: None
Wz: None

8.14 X-INVALID-NUMER-14 $Z(s) = \left(\frac{R_1}{C_1R_1s+1}, \infty, \infty, R_4, R_5 + \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{R_1R_4g_m + s(C_5R_1R_4R_5g_m - C_5R_1R_4)}{2R_1g_m + s^2(C_1C_5R_1R_4 + 2C_1C_5R_1R_5) + s(2C_1R_1 + 2C_5R_1R_4g_m + 2C_5R_1R_5g_m + 2C_5R_1 + C_5R_4 + 2C_5R_5) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}\sqrt{R_4+2R_5}\sqrt{R_1g_m+1}}{2C_1R_1+2C_5R_1R_4g_m+2C_5R_1R_5g_m+2C_5R_1+C_5R_4+2C_5R_5}$
wo: $\frac{\sqrt{2R_1g_m+2}}{\sqrt{C_1C_5R_1R_4+2C_1C_5R_1R_5}}$
bandwidth: $\frac{\sqrt{2}\sqrt{2R_1g_m+2}(2C_1R_1+2C_5R_1R_4g_m+2C_5R_1R_5g_m+2C_5R_1+C_5R_4+2C_5R_5)}{2\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}\sqrt{R_4+2R_5}\sqrt{R_1g_m+1}\sqrt{C_1C_5R_1R_4+2C_1C_5R_1R_5}}$
K-LP: $\frac{R_1R_4g_m}{2R_1g_m+2}$
K-HP: 0
K-BP: $\frac{C_5R_1R_4R_5g_m-C_5R_1R_4}{2C_1R_1+2C_5R_1R_4g_m+2C_5R_1R_5g_m+2C_5R_1+C_5R_4+2C_5R_5}$
Qz: None
Wz: None

8.15 X-INVALID-NUMER-15 $Z(s) = \left(\frac{R_1}{C_1R_1s+1}, \infty, \infty, \frac{1}{C_4s}, \frac{R_5}{C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{-C_5R_1R_5s + R_1R_5g_m - R_1}{2R_1g_m + s^2(2C_1C_4R_1R_5 + C_1C_5R_1R_5 + 2C_4C_5R_1R_5) + s(C_1R_1 + 2C_4R_1R_5g_m + 2C_4R_1 + 2C_4R_5 + 2C_5R_1R_5g_m + C_5R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{R_1}\sqrt{R_5}\sqrt{2R_1g_m+1}\sqrt{2C_1C_4+C_1C_5+2C_4C_5}}{C_1R_1+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5+2C_5R_1R_5g_m+C_5R_5}$
wo: $\frac{\sqrt{2R_1g_m+1}}{\sqrt{2C_1C_4R_1R_5+C_1C_5R_1R_5+2C_4C_5R_1R_5}}$
bandwidth: $\frac{C_1R_1+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5+2C_5R_1R_5g_m+C_5R_5}{\sqrt{R_1}\sqrt{R_5}\sqrt{2C_1C_4+C_1C_5+2C_4C_5}\sqrt{2C_1C_4R_1R_5+C_1C_5R_1R_5+2C_4C_5R_1R_5}}$
K-LP: $\frac{R_1R_5g_m-R_1}{2R_1g_m+1}$
K-HP: 0
K-BP: $-\frac{C_5R_1R_5}{C_1R_1+2C_4R_1R_5g_m+2C_4R_1+2C_4R_5+2C_5R_1R_5g_m+C_5R_5}$
Qz: None
Wz: None

8.16 X-INVALID-NUMER-16 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_4 s + R_1 R_4 g_m}{2 R_1 g_m + s^2 (2 C_1 C_4 R_1 R_4 + C_1 C_5 R_1 R_4 + 2 C_4 C_5 R_1 R_4) + s (2 C_1 R_1 + 2 C_4 R_1 R_4 g_m + 2 C_4 R_4 + 2 C_5 R_1 R_4 g_m + 2 C_5 R_1 + C_5 R_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{R_1}\sqrt{R_4}\sqrt{R_1 g_m + 1}\sqrt{2C_1 C_4 + C_1 C_5 + 2C_4 C_5}}{2C_1 R_1 + 2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4}$
 wo: $\frac{\sqrt{2R_1 g_m + 2}}{\sqrt{2C_1 C_4 R_1 R_4 + C_1 C_5 R_1 R_4 + 2C_4 C_5 R_1 R_4}}$
 bandwidth: $\frac{\sqrt{2}\sqrt{2R_1 g_m + 2}(2C_1 R_1 + 2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4)}{2\sqrt{R_1}\sqrt{R_4}\sqrt{R_1 g_m + 1}\sqrt{2C_1 C_4 + C_1 C_5 + 2C_4 C_5}\sqrt{2C_1 C_4 R_1 R_4 + C_1 C_5 R_1 R_4 + 2C_4 C_5 R_1 R_4}}$
 K-LP: $\frac{R_1 R_4 g_m}{2R_1 g_m + 2}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_1 R_4}{2C_1 R_1 + 2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4}$
 Qz: None
 Wz: None

8.17 X-INVALID-NUMER-17 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_4 R_5 s + R_1 R_4 R_5 g_m - R_1 R_4}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5 + s^2 (2 C_1 C_4 R_1 R_4 R_5 + C_1 C_5 R_1 R_4 R_5 + 2 C_4 C_5 R_1 R_4 R_5) + s (C_1 R_1 R_4 + 2 C_1 R_1 R_5 + 2 C_4 R_1 R_4 R_5 g_m + 2 C_4 R_1 R_4 + 2 C_4 R_4 R_5 + 2 C_5 R_1 R_4 R_5 g_m + 2 C_5 R_1 R_5 + C_5 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{R_1}\sqrt{R_4}\sqrt{R_5}\sqrt{2C_1 C_4 + C_1 C_5 + 2C_4 C_5}\sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{C_1 R_1 R_4 + 2C_1 R_1 R_5 + 2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5}$
 wo: $\frac{\sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{\sqrt{2C_1 C_4 R_1 R_4 R_5 + C_1 C_5 R_1 R_4 R_5 + 2C_4 C_5 R_1 R_4 R_5}}$
 bandwidth: $\frac{C_1 R_1 R_4 + 2C_1 R_1 R_5 + 2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5}{\sqrt{R_1}\sqrt{R_4}\sqrt{R_5}\sqrt{2C_1 C_4 + C_1 C_5 + 2C_4 C_5}\sqrt{2C_1 C_4 R_1 R_4 R_5 + C_1 C_5 R_1 R_4 R_5 + 2C_4 C_5 R_1 R_4 R_5}}$
 K-LP: $\frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_1 R_4 R_5}{C_1 R_1 R_4 + 2C_1 R_1 R_5 + 2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5}$
 Qz: None
 Wz: None

8.18 X-INVALID-NUMER-18 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s (C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4)}{2 R_1 g_m + s^2 (C_1 C_4 R_1 R_4 + 2 C_1 C_4 R_1 R_5) + s (C_1 R_1 + 2 C_4 R_1 R_4 g_m + 2 C_4 R_1 R_5 g_m + 2 C_4 R_1 + C_4 R_4 + 2 C_4 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4 + 2R_5}\sqrt{2R_1 g_m + 1}}{C_1 R_1 + 2C_4 R_1 R_4 g_m + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + C_4 R_4 + 2C_4 R_5}$
 wo: $\frac{\sqrt{2R_1 g_m + 1}}{\sqrt{C_1 C_4 R_1 R_4 + 2C_1 C_4 R_1 R_5}}$
 bandwidth: $\frac{C_1 R_1 + 2C_4 R_1 R_4 g_m + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + C_4 R_4 + 2C_4 R_5}{\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4 + 2R_5}\sqrt{C_1 C_4 R_1 R_4 + 2C_1 C_4 R_1 R_5}}$
 K-LP: $\frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1}$
 K-HP: 0
 K-BP: $\frac{C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4}{C_1 R_1 + 2C_4 R_1 R_4 g_m + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + C_4 R_4 + 2C_4 R_5}$
 Qz: None
 Wz: None

8.19 X-INVALID-NUMER-19 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_5 g_m + s (C_1 R_1 R_5 g_m - C_1 R_1) - 1}{2 g_m + s^2 (2 C_1 C_4 R_1 R_5 g_m + 2 C_1 C_4 R_1 + 2 C_1 C_4 R_5) + s (2 C_1 R_1 g_m + C_1 + 2 C_4 R_5 g_m + 2 C_4)}$$

Parameters:

Q: $\frac{2\sqrt{C_1}\sqrt{C_4}R_1R_5g_m\sqrt{\frac{g_m}{R_1R_5g_m+R_1+R_5}}+2\sqrt{C_1}\sqrt{C_4}R_1\sqrt{\frac{g_m}{R_1R_5g_m+R_1+R_5}}+2\sqrt{C_1}\sqrt{C_4}R_5\sqrt{\frac{g_m}{R_1R_5g_m+R_1+R_5}}}{2C_1R_1g_m+C_1+2C_4R_5g_m+2C_4}$
 wo: $\sqrt{\frac{g_m}{C_1C_4R_1R_5g_m+C_1C_4R_1+C_1C_4R_5}}$

$$\begin{aligned} \text{bandwidth: } & \frac{\sqrt{\frac{g_m}{C_1 C_4 R_1 R_5 g_m + C_1 C_4 R_1 + C_1 C_4 R_5}} (2C_1 R_1 g_m + C_1 + 2C_4 R_5 g_m + 2C_4)}{2\sqrt{C_1} \sqrt{C_4} R_1 R_5 g_m \sqrt{\frac{g_m}{R_1 R_5 g_m + R_1 + R_5}} + 2\sqrt{C_1} \sqrt{C_4} R_1 \sqrt{\frac{g_m}{R_1 R_5 g_m + R_1 + R_5}} + 2\sqrt{C_1} \sqrt{C_4} R_5 \sqrt{\frac{g_m}{R_1 R_5 g_m + R_1 + R_5}}} \\ \text{K-LP: } & \frac{R_5 g_m - 1}{2g_m} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_1 R_1 R_5 g_m - C_1 R_1}{2C_1 R_1 g_m + C_1 + 2C_4 R_5 g_m + 2C_4} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

$$\mathbf{8.20 \quad X-INVALID-NUMER-20} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s (C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4)}{2R_4 g_m + 2R_5 g_m + s^2 (2C_1 C_4 R_1 R_4 R_5 g_m + 2C_1 C_4 R_1 R_4 + 2C_1 C_4 R_4 R_5) + s (2C_1 R_1 R_4 g_m + 2C_1 R_1 R_5 g_m + 2C_1 R_1 + C_1 R_4 + 2C_1 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_4) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_1} \sqrt{C_4} R_1 \sqrt{R_4} R_5 g_m \sqrt{\frac{R_4 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_5 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}} + 2\sqrt{C_1} \sqrt{C_4} R_1 \sqrt{R_4} \sqrt{\frac{R_4 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_5 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}} + 2\sqrt{C_1} \sqrt{C_4} \sqrt{R_4} R_5 \sqrt{\frac{R_4 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_5 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}}}{2C_1 R_1 R_4 g_m + 2C_1 R_1 R_5 g_m + 2C_1 R_1 + C_1 R_4 + 2C_1 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_4} \\ \text{wo: } & \sqrt{\frac{R_4 g_m + R_5 g_m + 1}{C_1 C_4 R_1 R_4 R_5 g_m + C_1 C_4 R_1 R_4 + C_1 C_4 R_4 R_5}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{R_4 g_m + R_5 g_m + 1}{C_1 C_4 R_1 R_4 R_5 g_m + C_1 C_4 R_1 R_4 + C_1 C_4 R_4 R_5}} (2C_1 R_1 R_4 g_m + 2C_1 R_1 R_5 g_m + 2C_1 R_1 + C_1 R_4 + 2C_1 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_4)}{2\sqrt{C_1} \sqrt{C_4} R_1 \sqrt{R_4} R_5 g_m \sqrt{\frac{R_4 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_5 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}} + 2\sqrt{C_1} \sqrt{C_4} R_1 \sqrt{R_4} \sqrt{\frac{R_4 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_5 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}} + 2\sqrt{C_1} \sqrt{C_4} \sqrt{R_4} R_5 \sqrt{\frac{R_4 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{R_5 g_m}{R_1 R_5 g_m + R_1 + R_5} + \frac{1}{R_1 R_5 g_m + R_1 + R_5}}} \\ \text{K-LP: } & \frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + 2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4}{2C_1 R_1 R_4 g_m + 2C_1 R_1 R_5 g_m + 2C_1 R_1 + C_1 R_4 + 2C_1 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_4} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

9 X-INVALID-ORDER

$$\mathbf{9.1 \quad X-INVALID-ORDER-1} \quad Z(s) = (R_1, \quad \infty, \quad \infty, \quad R_4, \quad R_5, \quad \infty)$$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}$$

$$\mathbf{9.2 \quad X-INVALID-ORDER-2} \quad Z(s) = \left(R_1, \quad \infty, \quad \infty, \quad R_4, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 R_1 R_4 s + R_1 R_4 g_m}{2R_1 g_m + s (2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4) + 2}$$

$$\mathbf{9.3 \quad X-INVALID-ORDER-3} \quad Z(s) = \left(R_1, \quad \infty, \quad \infty, \quad R_4, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 R_1 R_4 R_5 s + R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s (2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5)}$$

$$\mathbf{9.4 \quad X-INVALID-ORDER-4} \quad Z(s) = \left(R_1, \quad \infty, \quad \infty, \quad R_4, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_1 R_4 g_m + s (C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4)}{2R_1 g_m + s (2C_5 R_1 R_4 g_m + 2C_5 R_1 R_5 g_m + 2C_5 R_1 + C_5 R_4 + 2C_5 R_5) + 2}$$

$$\mathbf{9.5 \quad X-INVALID-ORDER-5} \quad Z(s) = \left(R_1, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_1 R_5 g_m - R_1}{2R_1 g_m + s (2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5) + 1}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(R_1, \infty, \infty, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 s + R_1 g_m}{2C_4 C_5 R_1 s^2 + s(2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5)}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(R_1, \infty, \infty, \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 g_m + s(C_5 R_1 R_5 g_m - C_5 R_1)}{s^2(2C_4 C_5 R_1 R_5 g_m + 2C_4 C_5 R_1 + 2C_4 C_5 R_5) + s(2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5)}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(R_1, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s(2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5)}$$

9.9 X-INVALID-ORDER-9 $Z(s) = \left(R_1, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s(C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4)}{2R_1 g_m + s(2C_4 R_1 R_4 g_m + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + C_4 R_4 + 2C_4 R_5) + 1}$$

9.10 X-INVALID-ORDER-10 $Z(s) = \left(R_1, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 R_1 R_4 s^2 + R_1 g_m + s(C_4 R_1 R_4 g_m - C_5 R_1)}{s^2(2C_4 C_5 R_1 R_4 g_m + 2C_4 C_5 R_1 + C_4 C_5 R_4) + s(2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5)}$$

9.11 X-INVALID-ORDER-11 $Z(s) = \left(R_1, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 g_m + s^2(C_4 C_5 R_1 R_4 R_5 g_m - C_4 C_5 R_1 R_4) + s(C_4 R_1 R_4 g_m + C_5 R_1 R_5 g_m - C_5 R_1)}{s^2(2C_4 C_5 R_1 R_4 g_m + 2C_4 C_5 R_1 R_5 g_m + 2C_4 C_5 R_1 + C_4 C_5 R_4 + 2C_4 C_5 R_5) + s(2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5)}$$

9.12 X-INVALID-ORDER-12 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, R_4, R_5, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + s(C_1 R_4 + 2C_1 R_5) + 2}$$

9.13 X-INVALID-ORDER-13 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 s + g_m}{s^2(2C_1 C_4 + C_1 C_5 + 2C_4 C_5) + s(2C_4 g_m + 2C_5 g_m)}$$

9.14 X-INVALID-ORDER-14 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{g_m + s(C_5 R_5 g_m - C_5)}{2C_1 C_4 C_5 R_5 s^3 + s^2(2C_1 C_4 + C_1 C_5 + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s(2C_4 g_m + 2C_5 g_m)}$$

9.15 X-INVALID-ORDER-15 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_4 g_m + s(C_5 R_4 R_5 g_m - C_5 R_4)}{2C_1 C_4 C_5 R_4 R_5 s^3 + 2g_m + s^2(2C_1 C_4 R_4 + C_1 C_5 R_4 + 2C_1 C_5 R_5 + 2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_4) + s(2C_1 + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5)}$$

$$\mathbf{9.16 \quad X-INVALID-ORDER-16} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_4 s^2 + g_m + s(C_4 R_4 g_m - C_5)}{C_1 C_4 C_5 R_4 s^3 + s^2(2C_1 C_4 + C_1 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5) + s(2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.17 \quad X-INVALID-ORDER-17} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_4 R_5 s^2 + R_5 g_m + s(C_4 R_4 R_5 g_m - C_4 R_4 - C_5 R_5) - 1}{C_1 C_4 C_5 R_4 R_5 s^3 + 2g_m + s^2(C_1 C_4 R_4 + 2C_1 C_4 R_5 + C_1 C_5 R_5 + 2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_5) + s(C_1 + 2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m)}$$

$$\mathbf{9.18 \quad X-INVALID-ORDER-18} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{g_m + s^2(C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4) + s(C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{s^3(C_1 C_4 C_5 R_4 + 2C_1 C_4 C_5 R_5) + s^2(2C_1 C_4 + C_1 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s(2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.19 \quad X-INVALID-ORDER-19} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s(C_1 R_1 R_4 + 2C_1 R_1 R_5)}$$

$$\mathbf{9.20 \quad X-INVALID-ORDER-20} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_5 R_1 s + R_1 g_m}{s^2(2C_1 C_4 R_1 + C_1 C_5 R_1 + 2C_4 C_5 R_1) + s(2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.21 \quad X-INVALID-ORDER-21} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s(C_5 R_1 R_5 g_m - C_5 R_1)}{2C_1 C_4 C_5 R_1 R_5 s^3 + s^2(2C_1 C_4 R_1 + C_1 C_5 R_1 + 2C_4 C_5 R_1 R_5 g_m + 2C_4 C_5 R_1 + 2C_4 C_5 R_5) + s(2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.22 \quad X-INVALID-ORDER-22} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 R_4 g_m + s(C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4)}{2C_1 C_4 C_5 R_1 R_4 R_5 s^3 + 2R_1 g_m + s^2(2C_1 C_4 R_1 R_4 + C_1 C_5 R_1 R_4 + 2C_1 C_5 R_1 R_5 + 2C_4 C_5 R_1 R_4 R_5 g_m + 2C_4 C_5 R_1 R_4 + 2C_4 C_5 R_4 R_5) + s(2C_1 R_1 + 2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 R_5 g_m + 2C_5 R_1 + C_5 R_4 + 2C_5 R_5) + 2}$$

$$\mathbf{9.23 \quad X-INVALID-ORDER-23} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_1 R_4 s^2 + R_1 g_m + s(C_4 R_1 R_4 g_m - C_5 R_1)}{C_1 C_4 C_5 R_1 R_4 s^3 + s^2(2C_1 C_4 R_1 + C_1 C_5 R_1 + 2C_4 C_5 R_1 R_4 g_m + 2C_4 C_5 R_1 + C_4 C_5 R_4) + s(2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.24 \quad X-INVALID-ORDER-24} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_1 R_4 R_5 s^2 + R_1 R_5 g_m - R_1 + s(C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4 - C_5 R_1 R_5)}{C_1 C_4 C_5 R_1 R_4 R_5 s^3 + 2R_1 g_m + s^2(C_1 C_4 R_1 R_4 + 2C_1 C_4 R_1 R_5 + C_1 C_5 R_1 R_5 + 2C_4 C_5 R_1 R_4 R_5 g_m + 2C_4 C_5 R_1 R_5 + C_4 C_5 R_4 R_5) + s(C_1 R_1 + 2C_4 R_1 R_4 g_m + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + C_4 R_4 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

$$\mathbf{9.25 \quad X-INVALID-ORDER-25} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s^2(C_4 C_5 R_1 R_4 R_5 g_m - C_4 C_5 R_1 R_4) + s(C_4 R_1 R_4 g_m + C_5 R_1 R_5 g_m - C_5 R_1)}{s^3(C_1 C_4 C_5 R_1 R_4 + 2C_1 C_4 C_5 R_1 R_5) + s^2(2C_1 C_4 R_1 + C_1 C_5 R_1 + 2C_4 C_5 R_1 R_4 g_m + 2C_4 C_5 R_1 R_5 g_m + 2C_4 C_5 R_1 + C_4 C_5 R_4 + 2C_4 C_5 R_5) + s(2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5)}$$

9.26 X-INVALID-ORDER-26 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, R_4, R_5, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s (C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4)}{2 R_4 g_m + 2 R_5 g_m + s (2 C_1 R_1 R_4 g_m + 2 C_1 R_1 R_5 g_m + 2 C_1 R_1 + C_1 R_4 + 2 C_1 R_5) + 2}$$

9.27 X-INVALID-ORDER-27 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 R_1 s^2 + g_m + s (C_1 R_1 g_m - C_5)}{2 C_1 C_4 C_5 R_1 s^3 + s^2 (2 C_1 C_4 R_1 g_m + 2 C_1 C_4 + 2 C_1 C_5 R_1 g_m + C_1 C_5 + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

9.28 X-INVALID-ORDER-28 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 R_1 R_5 s^2 + R_5 g_m + s (C_1 R_1 R_5 g_m - C_1 R_1 - C_5 R_5) - 1}{2 C_1 C_4 C_5 R_1 R_5 s^3 + 2 g_m + s^2 (2 C_1 C_4 R_1 R_5 g_m + 2 C_1 C_4 R_1 + 2 C_1 C_4 R_5 + 2 C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_5 + 2 C_4 C_5 R_5) + s (2 C_1 R_1 g_m + C_1 + 2 C_4 R_5 g_m + 2 C_4 + 2 C_5 R_5 g_m)}$$

9.29 X-INVALID-ORDER-29 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{g_m + s^2 (C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1) + s (C_1 R_1 g_m + C_5 R_5 g_m - C_5)}{s^3 (2 C_1 C_4 C_5 R_1 R_5 g_m + 2 C_1 C_4 C_5 R_1 + 2 C_1 C_4 C_5 R_5) + s^2 (2 C_1 C_4 R_1 g_m + 2 C_1 C_4 + 2 C_1 C_5 R_1 g_m + C_1 C_5 + 2 C_4 C_5 R_5 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

9.30 X-INVALID-ORDER-30 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 R_1 R_4 s^2 + R_4 g_m + s (C_1 R_1 R_4 g_m - C_5 R_4)}{2 C_1 C_4 C_5 R_1 R_4 s^3 + 2 g_m + s^2 (2 C_1 C_4 R_1 R_4 g_m + 2 C_1 C_4 R_4 + 2 C_1 C_5 R_1 R_4 g_m + 2 C_1 C_5 R_1 + C_1 C_5 R_4 + 2 C_4 C_5 R_4) + s (2 C_1 R_1 g_m + 2 C_1 + 2 C_4 R_4 g_m + 2 C_5 R_4 g_m + 2 C_5)}$$

9.31 X-INVALID-ORDER-31 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 R_1 R_4 R_5 s^2 + R_4 R_5 g_m - R_4 + s (C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4 - C_5 R_4 R_5)}{2 C_1 C_4 C_5 R_1 R_4 R_5 s^3 + 2 R_4 g_m + 2 R_5 g_m + s^2 (2 C_1 C_4 R_1 R_4 R_5 g_m + 2 C_1 C_4 R_1 R_4 + 2 C_1 C_4 R_4 R_5 + 2 C_1 C_5 R_1 R_4 R_5 g_m + 2 C_1 C_5 R_1 R_5 + C_1 C_5 R_4 R_5 + 2 C_4 C_5 R_4 R_5) + s (2 C_1 R_1 R_4 g_m + 2 C_1 R_1 R_5 g_m + 2 C_1 R_1 + C_1 R_4 + 2 C_1 R_5 + 2 C_4 R_4 R_5 g_m + 2 C_4 R_4 + 2 C_5 R_4 R_5 g_m + 2 C_5 R_4 R_5)}$$

9.32 X-INVALID-ORDER-32 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_4 g_m + s^2 (C_1 C_5 R_1 R_4 R_5 g_m - C_1 C_5 R_1 R_4) + s (C_1 R_1 R_4 g_m + C_5 R_4 R_5 g_m - C_5 R_4)}{2 g_m + s^3 (2 C_1 C_4 C_5 R_1 R_4 R_5 g_m + 2 C_1 C_4 C_5 R_1 R_4 + 2 C_1 C_4 C_5 R_4 R_5) + s^2 (2 C_1 C_4 R_1 R_4 g_m + 2 C_1 C_4 R_4 + 2 C_1 C_5 R_1 R_4 g_m + 2 C_1 C_5 R_1 R_5 g_m + 2 C_1 C_5 R_1 + C_1 C_5 R_4 + 2 C_1 C_5 R_5 + 2 C_4 C_5 R_4 R_5 g_m + 2 C_4 C_5 R_4) + s (2 C_1 R_1 g_m + 2 C_1 + 2 C_4 R_4 g_m + 2 C_5 R_4 g_m + 2 C_5 R_5 g_m)}$$

9.33 X-INVALID-ORDER-33 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 R_1 R_4 s^3 + g_m + s^2 (C_1 C_4 R_1 R_4 g_m - C_1 C_5 R_1 - C_4 C_5 R_4) + s (C_1 R_1 g_m + C_4 R_4 g_m - C_5)}{s^3 (2 C_1 C_4 C_5 R_1 R_4 g_m + 2 C_1 C_4 C_5 R_1 + C_1 C_4 C_5 R_4) + s^2 (2 C_1 C_4 R_1 g_m + 2 C_1 C_4 + 2 C_1 C_5 R_1 g_m + C_1 C_5 + 2 C_4 C_5 R_4 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

9.34 X-INVALID-ORDER-34 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 R_1 R_4 R_5 s^3 + R_5 g_m + s^2 (C_1 C_4 R_1 R_4 R_5 g_m - C_1 C_4 R_1 R_4 - C_1 C_5 R_1 R_5 - C_4 C_5 R_4 R_5) + s (C_1 R_1 R_5 g_m - C_1 R_1 + C_4 R_4 R_5 g_m - C_4 R_4 - C_5 R_5) - 1}{2 g_m + s^3 (2 C_1 C_4 C_5 R_1 R_4 R_5 g_m + 2 C_1 C_4 C_5 R_1 R_5 + C_1 C_4 C_5 R_4 R_5) + s^2 (2 C_1 C_4 R_1 R_4 g_m + 2 C_1 C_4 R_1 R_5 g_m + 2 C_1 C_4 R_1 + C_1 C_4 R_4 + 2 C_1 C_4 R_5 + 2 C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_5 + 2 C_4 C_5 R_4 R_5 g_m + 2 C_4 C_5 R_5) + s (2 C_1 R_1 g_m + C_1 + 2 C_4 R_4 g_m + 2 C_4 R_5 g_m + 2 C_4 + 2 C_5 R_5)}$$

9.35 X-INVALID-ORDER-35 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{g_m + s^3 (C_1 C_4 C_5 R_1 R_4 R_5 g_m - C_1 C_4 C_5 R_1 R_4) + s^2 (C_1 C_4 R_1 R_4 g_m + C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1 + C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4) + s (C_1 R_1 g_m + C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{s^3 (2 C_1 C_4 C_5 R_1 R_4 g_m + 2 C_1 C_4 C_5 R_1 R_5 g_m + 2 C_1 C_4 C_5 R_1 + C_1 C_4 C_5 R_4 + 2 C_1 C_4 C_5 R_5) + s^2 (2 C_1 C_4 R_1 g_m + 2 C_1 C_4 + 2 C_1 C_5 R_1 g_m + C_1 C_5 + 2 C_4 C_5 R_4 g_m + 2 C_4 C_5 R_5 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(R_1, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 R_1 R_4 R_5 s^2 + R_1 R_5 g_m - R_1 + s(C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4 - C_5 R_1 R_5)}{2R_1 g_m + s^2(2C_4 C_5 R_1 R_4 R_5 g_m + 2C_4 C_5 R_1 R_5 + C_4 C_5 R_4 R_5) + s(2C_4 R_1 R_4 g_m + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + C_4 R_4 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_4}\sqrt{C_5}R_1R_4\sqrt{R_5}g_m\sqrt{\frac{2R_1g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}+2\sqrt{C_4}\sqrt{C_5}R_1\sqrt{R_5}\sqrt{\frac{2R_1g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}+\sqrt{C_4}\sqrt{C_5}R_4\sqrt{R_5}\sqrt{\frac{2R_1g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}}{2C_4R_1R_4g_m+2C_4R_1R_5g_m+2C_4R_1+C_4R_4+2C_4R_5+2C_5R_1R_5g_m+C_5R_5} \\ \text{wo: } & \sqrt{\frac{2R_1g_m+1}{2C_4C_5R_1R_4R_5g_m+2C_4C_5R_1R_5+C_4C_5R_4R_5}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{2R_1g_m+1}{2C_4C_5R_1R_4R_5g_m+2C_4C_5R_1R_5+C_4C_5R_4R_5}}(2C_4R_1R_4g_m+2C_4R_1R_5g_m+2C_4R_1+C_4R_4+2C_4R_5+2C_5R_1R_5g_m+C_5R_5)}{2\sqrt{C_4}\sqrt{C_5}R_1R_4\sqrt{R_5}g_m\sqrt{\frac{2R_1g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}+2\sqrt{C_4}\sqrt{C_5}R_1\sqrt{R_5}\sqrt{\frac{2R_1g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}+\sqrt{C_4}\sqrt{C_5}R_4\sqrt{R_5}\sqrt{\frac{2R_1g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}} \\ \text{K-LP: } & \frac{R_1R_5g_m-R_1}{2R_1g_m+1} \\ \text{K-HP: } & -\frac{R_1R_4}{2R_1R_4g_m+2R_1+R_4} \\ \text{K-BP: } & \frac{C_4R_1R_4R_5g_m-C_4R_1R_4-C_5R_1R_5}{2C_4R_1R_4g_m+2C_4R_1R_5g_m+2C_4R_1+C_4R_4+2C_4R_5+2C_5R_1R_5g_m+C_5R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \frac{\sqrt{-R_5g_m+1}}{\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}\sqrt{R_5}} \end{aligned}$$

10.2 X-INVALID-WZ-2 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, R_4, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 R_1 R_4 s^2 + R_4 g_m + s(C_1 R_1 R_4 g_m - C_5 R_4)}{2g_m + s^2(2C_1 C_5 R_1 R_4 g_m + 2C_1 C_5 R_1 + C_1 C_5 R_4) + s(2C_1 R_1 g_m + 2C_1 + 2C_5 R_4 g_m + 2C_5)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1R_4g_m\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1+R_4}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1+R_4}}+\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_4\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1+R_4}}}{2C_1R_1g_m+2C_1+2C_5R_4g_m+2C_5} \\ \text{wo: } & \sqrt{2}\sqrt{\frac{g_m}{2C_1C_5R_1R_4g_m+2C_1C_5R_1+C_1C_5R_4}} \\ \text{bandwidth: } & \frac{\sqrt{2}\sqrt{\frac{g_m}{2C_1C_5R_1R_4g_m+2C_1C_5R_1+C_1C_5R_4}}(2C_1R_1g_m+2C_1+2C_5R_4g_m+2C_5)}{2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1R_4g_m\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1+R_4}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1+R_4}}+\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_4\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1+R_4}}} \\ \text{K-LP: } & \frac{R_4}{2} \\ \text{K-HP: } & -\frac{R_1R_4}{2R_1R_4g_m+2R_1+R_4} \\ \text{K-BP: } & \frac{C_1R_1R_4g_m-C_5R_4}{2C_1R_1g_m+2C_1+2C_5R_4g_m+2C_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \frac{\sqrt{-g_m}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}} \end{aligned}$$

10.3 X-INVALID-WZ-3 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, R_4, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 R_1 R_4 R_5 s^2 + R_4 R_5 g_m - R_4 + s(C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4 - C_5 R_4 R_5)}{2R_4 g_m + 2R_5 g_m + s^2(2C_1 C_5 R_1 R_4 R_5 g_m + 2C_1 C_5 R_1 R_5 + C_1 C_5 R_4 R_5) + s(2C_1 R_1 R_4 g_m + 2C_1 R_1 R_5 g_m + 2C_1 R_1 + C_1 R_4 + 2C_1 R_5 + 2C_5 R_4 R_5 g_m + 2C_5 R_5) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1R_4\sqrt{R_5}g_m\sqrt{\frac{R_4g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{R_5g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1\sqrt{R_5}\sqrt{\frac{R_4g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{R_5g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}+\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_4\sqrt{R_5}\sqrt{\frac{R_4g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{R_5g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}}{2C_1R_1R_4g_m+2C_1R_1R_5g_m+2C_1R_1+C_1R_4+2C_1R_5+2C_5R_4R_5g_m+2C_5R_5} \\ \text{wo: } & \sqrt{\frac{2R_4g_m+2R_5g_m+2}{2C_1C_5R_1R_4R_5g_m+2C_1C_5R_1R_5+C_1C_5R_4R_5}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{2R_4g_m+2R_5g_m+2}{2C_1C_5R_1R_4R_5g_m+2C_1C_5R_1R_5+C_1C_5R_4R_5}}(2C_1R_1R_4g_m+2C_1R_1R_5g_m+2C_1R_1+C_1R_4+2C_1R_5+2C_5R_4R_5g_m+2C_5R_5)}{2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1R_4\sqrt{R_5}g_m\sqrt{\frac{R_4g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{R_5g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1\sqrt{R_5}\sqrt{\frac{R_4g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{R_5g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}+\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_4\sqrt{R_5}\sqrt{\frac{R_4g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{R_5g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}} \\ \text{K-LP: } & \frac{R_4R_5g_m-R_4}{2R_4g_m+2R_5g_m+2} \\ \text{K-HP: } & -\frac{R_1R_4}{2R_1R_4g_m+2R_1+R_4} \\ \text{K-BP: } & \frac{C_1R_1R_4R_5g_m-C_1R_1R_4-C_5R_4R_5}{2C_1R_1R_4g_m+2C_1R_1R_5g_m+2C_1R_1+C_1R_4+2C_1R_5+2C_5R_4R_5g_m+2C_5R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \frac{\sqrt{-R_5g_m+1}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}\sqrt{R_5}} \end{aligned}$$

10.4 X-INVALID-WZ-4 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, R_4, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_4 g_m + s^2 (C_1 C_5 R_1 R_4 R_5 g_m - C_1 C_5 R_1 R_4) + s (C_1 R_1 R_4 g_m + C_5 R_4 R_5 g_m - C_5 R_4)}{2g_m + s^2 (2C_1 C_5 R_1 R_4 g_m + 2C_1 C_5 R_1 R_5 g_m + 2C_1 C_5 R_1 + C_1 C_5 R_4 + 2C_1 C_5 R_5) + s (2C_1 R_1 g_m + 2C_1 + 2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5)}$$

Parameters:

$$\text{Q: } \frac{2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1R_4g_m\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1R_5g_m\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_4\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_5\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}}{2C_1R_1g_m+2C_1+2C_5R_4g_m+2C_5R_5g_m+2C_5}$$

$$\text{wo: } \sqrt{2}\sqrt{\frac{g_m}{2C_1C_5R_1R_4g_m+2C_1C_5R_1R_5g_m+2C_1C_5R_1+C_1C_5R_4+2C_1C_5R_5}}$$

$$\text{bandwidth: } \frac{\sqrt{2}\sqrt{\frac{g_m}{2C_1C_5R_1R_4g_m+2C_1C_5R_1R_5g_m+2C_1C_5R_1+C_1C_5R_4+2C_1C_5R_5}}(2C_1R_1g_m+2C_1+2C_5R_4g_m+2C_5R_5g_m+2C_5)}{2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1R_4g_m\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1R_5g_m\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_4\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_5\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}}$$

$$\text{K-LP: } \frac{R_4}{2}$$

$$\text{K-HP: } \frac{R_1R_4R_5g_m-R_1R_4}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}$$

$$\text{K-BP: } \frac{C_1R_1R_4g_m+C_5R_4R_5g_m-C_5R_4}{2C_1R_1g_m+2C_1+2C_5R_4g_m+2C_5R_5g_m+2C_5}$$

$$\text{Qz: None}$$

$$\text{Wz: } \sqrt{\frac{g_m}{C_1C_5R_1R_5g_m-C_1C_5R_1}}$$

10.5 X-INVALID-WZ-5 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_5 g_m + s^2 (C_1 C_4 R_1 R_4 R_5 g_m - C_1 C_4 R_1 R_4) + s (C_1 R_1 R_5 g_m - C_1 R_1 + C_4 R_4 R_5 g_m - C_4 R_4) - 1}{2g_m + s^2 (2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_1 R_5 g_m + 2C_1 C_4 R_1 + C_1 C_4 R_4 + 2C_1 C_4 R_5) + s (2C_1 R_1 g_m + C_1 + 2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4)}$$

Parameters:

$$\text{Q: } \frac{2\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1R_4g_m\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1R_5g_m\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_4\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_5\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}}{2C_1R_1g_m+C_1+2C_4R_4g_m+2C_4R_5g_m+2C_4}$$

$$\text{wo: } \sqrt{2}\sqrt{\frac{g_m}{2C_1C_4R_1R_4g_m+2C_1C_4R_1R_5g_m+2C_1C_4R_1+C_1C_4R_4+2C_1C_4R_5}}$$

$$\text{bandwidth: } \frac{\sqrt{2}\sqrt{\frac{g_m}{2C_1C_4R_1R_4g_m+2C_1C_4R_1R_5g_m+2C_1C_4R_1+C_1C_4R_4+2C_1C_4R_5}}(2C_1R_1g_m+C_1+2C_4R_4g_m+2C_4R_5g_m+2C_4)}{2\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1R_4g_m\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1R_5g_m\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_4\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_5\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}}$$

$$\text{K-LP: } \frac{R_5g_m-1}{2g_m}$$

$$\text{K-HP: } \frac{R_1R_4R_5g_m-R_1R_4}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}$$

$$\text{K-BP: } \frac{C_1R_1R_5g_m-C_1R_1+C_4R_4R_5g_m-C_4R_4}{2C_1R_1g_m+C_1+2C_4R_4g_m+2C_4R_5g_m+2C_4}$$

$$\text{Qz: None}$$

$$\text{Wz: } \frac{1}{\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4}}$$

11 X-PolynomialError